

A Retraction-Free EXTRA Method for Decentralized Optimization on the Stiefel Manifold

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Abstract. Decentralized optimization provides a fundamental framework for large-scale learning and signal processing with distributed data. We study decentralized optimization with orthogonality constraints on the Stiefel manifold and propose RF-EXTRA, a distributed retraction-free primal-dual method on static undirected networks. The method combines an approximate gradient mapping for orthogonality-constrained optimization with an EXTRA-based decentralized recursion, thereby avoiding explicit retractions while preserving a simple communication pattern. On the theoretical side, the analysis considers both the averaged iterate and the joint error $(\mathbf{X}_k - \bar{\mathbf{X}}_k, \mathbf{s}_k - \bar{\mathbf{s}}_k)$, and establishes a contractive recursion for the joint error under an equivalent norm. This recursion is then used to compare the different terms in the descent lemma and finally derive an exact $\mathcal{O}(1/K)$ convergence rate for the averaged iterate under explicit parameter conditions. Experiments on PCA and low-rank matrix completion show that RF-EXTRA compares favorably with the reported decentralized baselines and exhibits strong communication efficiency on the tested tasks on the Stiefel manifold.

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Key words: decentralized optimization, Stiefel manifold, retraction-free method, EXTRA, non-convex optimization

1 Introduction

Decentralized optimization [1,2] studies how a network of agents cooperatively solves a global problem using local data and local communication. It has become a fundamental framework in large-scale learning [3], signal processing [4,5], and control [1], where data are naturally distributed and centralized aggregation is either costly or undesirable. In its classical Euclidean form, the goal is to combine computational scalability with communication efficiency while preserving the solution quality of the underlying centralized model.

Many decentralized problems are not purely Euclidean. In distributed principal component analysis, subspace estimation, and related matrix-learning tasks, the decision

variable is required to have orthonormal columns, and the optimization therefore takes place on the Stiefel manifold

$$\text{St}(d, r) = \{X \in \mathbb{R}^{d \times r} : X^\top X = I_r\}.$$

Such orthogonality constraints also arise in deep models with orthogonal layers [6], normalization mechanisms [7], and low-rank fine-tuning schemes [8]. Accordingly, decentralized optimization on the Stiefel manifold has become an important problem in decentralized learning and matrix optimization.

A central difficulty is that decentralized primal-dual correction mechanisms and manifold feasibility mechanisms are driven by different principles. Methods in the EXTRA family are analyzed as primal-dual type recursions, whereas manifold methods typically rely on tangent projections and retractions to preserve feasibility. Once EXTRA is brought into a nonconvex problem on the Stiefel manifold, combining these two ingredients is essentially equivalent to analyzing a primal-dual algorithm together with nonconvex manifold constraints and geometry in a single recursion. The convergence analysis is hence highly nontrivial.

This paper studies the consensus-constrained optimization problem on the Stiefel manifold

$$\min_{X \in \text{St}(d, r)} F(X) := \frac{1}{n} \sum_{i=1}^n f_i(X),$$

where each agent $i \in \{1, \dots, n\}$ has access only to its local smooth objective f_i and communicates over a fixed connected undirected graph. Our aim is to develop a decentralized method that preserves the correction mechanism of EXTRA while using a retraction-free update on the Stiefel manifold. This direction is algorithmically attractive because modern GPUs are highly optimized for basic linear algebra subprograms (BLAS) [9], whereas retraction operations such as repeated QR or polar orthogonalization are typically less GPU-friendly than simple matrix multiplications and additions.

1.1 Related Work

1.1.1 Primal methods and primal-dual methods in decentralized optimization

The development of decentralized optimization began with primal methods such as decentralized gradient descent (DGD) [2] and decentralized SGD type variants for stochastic objectives [10, 11], which established the basic consensus optimization framework but generally suffer from steady state bias under constant stepsizes. To remove that bias, later primal-dual or correction based methods introduced auxiliary states that couple consensus and optimization more tightly. Representative milestones include gradient tracking and DGT type methods [12–14], EXTRA [4], exact diffusion [5], NIDS [15], and prox-PDA [16]. These methods clarified how exactness can be preserved over static networks while maintaining low per-iteration communication cost.

The surrounding Euclidean literature further expanded this line through refined analyses for heterogeneous and nonconvex settings [3, 17, 18], compressed communication [19–23], time varying and directed networks [24, 25], local step variants [26], and graph design viewpoints [27]. These Euclidean results provide the algorithmic backbone for our method, since RF-EXTRA keeps EXTRA itself while adapting it to optimization on the Stiefel manifold.

1.1.2 Distributed optimization on the Stiefel manifold

For orthogonality-constrained problems, decentralized Riemannian methods form the natural baseline. DRSGD [28] studies decentralized Riemannian gradient descent on the Stiefel manifold and shows that distributed PCA type tasks can be handled directly in the manifold setting. More recently, DPRGD [29] and DPRGT [30] study projected decentralized Riemannian gradient updates and decentralized proximal gradient tracking on Riemannian manifolds, respectively; together they provide our main projection based and tracking based baselines. Riemannian EXTRA (REXTRA) [31] further demonstrates that EXTRA type primal-dual correction can be extended from Euclidean space to compact submanifolds, but still through retraction or projection based updates.

Related works further considered quantization and compression [32, 33], stochastic recursive momentum [34], conjugate gradient and natural gradient directions [35, 36], local linear consensus behavior on the Stiefel manifold [37], manifold consensus formulations [38, 39], and decentralized retraction-free optimization on the Stiefel manifold [40]. However, an EXTRA type correction method in this retraction-free regime is still unavailable.

1.1.3 Retraction-free optimization with orthogonality constraints

A separate line of work studies orthogonality-constrained optimization without performing explicit retractions at every iteration. DESTINY [41] introduced an approximate gradient mapping for orthogonality-constrained optimization through an approximate augmented Lagrangian formulation on the Stiefel manifold, where orthogonality is controlled by a penalty-based surrogate instead of being enforced by explicit retractions at every step. Early centralized retraction-free methods also showed that one can reduce the cost of repeated QR or polar decompositions while still controlling feasibility drift [42, 43]. This perspective is particularly attractive for large matrix problems, where orthogonalization can dominate the local computational budget.

More recent developments include infeasible methods with local linear convergence guarantees [44], ambient-space reformulations that dissolve hard constraints into smooth penalty structures [45], and preconditioned schemes motivated by LoRA type large-model fine-tuning [8]. Related applications in orthogonal weight normalization and multi-Stiefel learning further illustrate the practical importance of low-cost updates for orthogonality-constrained problems [6, 7]. The present work draws on this line to construct the local geometric step, and it combines that step with the decentralized EXTRA correction mechanism discussed above.

1.2 Contribution and Organization

This paper combines decentralized primal-dual correction, distributed optimization on the Stiefel manifold, and retraction-free orthogonality control in a single scheme. The key idea is to embed an approximate gradient mapping for orthogonality-constrained optimization into an EXTRA recursion.

Our contributions are summarized as follows.

- We design and implement RF-EXTRA, a distributed retraction-free EXTRA algorithm that combines an approximate gradient mapping for orthogonality-constrained optimization with an EXTRA-based decentralized recursion on the Stiefel manifold, and we validate this design through a systematic numerical study on PCA and LRMC over the Stiefel manifold.
- On the theory side, we show that the joint-error variables $(\mathbf{X}_k - \bar{\mathbf{X}}_k, \mathbf{s}_k - \bar{\mathbf{s}}_k)$ remain in a controllable region and satisfy a contractive recursion under an equivalent norm. This allows us to establish a summed joint-error bound, compare the different terms in the descent analysis of the original objective function, and finally derive an exact $\mathcal{O}(1/K)$ convergence rate under a constant stepsize. In contrast to existing retraction-based EXTRA-type methods such as REXTRA [31] and existing retraction-free schemes without EXTRA correction, our analysis must couple retraction-free feasibility control with an EXTRA-type inexact primal-dual recursion; this combination is nontrivial because of the manifold nonconvexity, especially under a constant stepsize, as also noted in [31].

This article is organized as follows. Section 2 presents RF-EXTRA together with the approximate gradient mapping for orthogonality-constrained optimization. Section 3 first states the standing assumptions and the averaged-sequence ingredients in Lemma 3.1, then introduces the equivalent-norm lemma and the joint-error recursion, next proves neighborhood propagation in Lemma 3.4, and finally establishes the exact $\mathcal{O}(1/K)$ convergence theorem.

1.3 Notation

Throughout the theoretical development, we use n for the number of agents and $X \in \mathbb{R}^{d \times r}$ for the matrix variable on the Stiefel manifold. We write

$$\mathbf{X}_k = [X_{1,k}^\top, \dots, X_{n,k}^\top]^\top, \quad \mathbf{s}_k = [s_{1,k}^\top, \dots, s_{n,k}^\top]^\top$$

for the stacked primal and auxiliary variables, and

$$\bar{X}_k := \frac{1}{n} \sum_{i=1}^n X_{i,k}, \quad \bar{s}_k := \frac{1}{n} \sum_{i=1}^n s_{i,k}$$

for their averages. The consensus projector is

$$J := \frac{1}{n} \mathbf{1}_n \mathbf{1}_n^\top \otimes I_d,$$

and we use

$$\bar{\mathbf{X}}_k := (\mathbf{1}_n \otimes I_d) \bar{X}_k, \quad \bar{\mathbf{s}}_k := (\mathbf{1}_n \otimes I_d) \bar{s}_k.$$

We also use the lifted mixing matrices

$$\mathbf{W} := W \otimes I_d, \quad \mathbf{V} := V \otimes I_d,$$

so the stacked block recursions are written directly in terms of $\mathbf{W}$ and $\mathbf{V}$. The stacked Euclidean gradients are denoted by

$$\nabla \mathbf{f}(\mathbf{X}) := [\nabla f_1(X_1)^\top, \dots, \nabla f_n(X_n)^\top]^\top.$$

The joint error is defined by

$$\mathbf{p}_k := \begin{bmatrix} \mathbf{X}_k - \bar{\mathbf{X}}_k \\ \mathbf{s}_k - \bar{\mathbf{s}}_k \end{bmatrix}.$$

Finally, $\|\cdot\|_F$ denotes the Frobenius norm.

2 A Retraction-Free EXTRA Method for Decentralized Optimization

We consider the decentralized optimization problem on the Stiefel manifold introduced in Section 1. Each agent $i \in \{1, \dots, n\}$ stores a local variable $X_{i,k} \in \mathbb{R}^{d \times r}$ and has access only to its local objective f_i . Let $W \in \mathbb{R}^{n \times n}$ be a symmetric doubly stochastic mixing matrix associated with a connected undirected graph, and let

$$V = \theta I_n + (1 - \theta)W, \quad \theta \in (0, 1/2].$$

We use the lifted mixing matrices $\mathbf{W}$ and $\mathbf{V}$ introduced in the notation section. Our construction starts from the Euclidean correction philosophy of EXTRA and then replaces the Euclidean gradient with a surrogate compatible with the geometry of the Stiefel manifold.

For unconstrained consensus optimization, EXTRA corrects the steady state bias of decentralized gradient descent by coupling the primal recursion with a memory term. An equivalent two-state representation of the Euclidean mechanism can be written as

$$\mathbf{X}_{k+1} = \mathbf{W}\mathbf{X}_k + \mathbf{s}_k, \tag{2.1}$$

$$\mathbf{s}_{k+1} = \mathbf{s}_k + (\mathbf{W} - \mathbf{V})\mathbf{X}_k - \alpha (\nabla \mathbf{f}(\mathbf{X}_{k+1}) - \nabla \mathbf{f}(\mathbf{X}_k)), \tag{2.2}$$

where $\nabla f(\mathbf{X}_k)$ is defined in the notation section. This representation makes the role of the auxiliary state explicit: the matrix $\mathbf{W} - \mathbf{V}$ handles the consensus correction, while the gradient difference drives descent without introducing a persistent bias under heterogeneity [4, 5]. Our goal is to retain this correction architecture in the setting of the Stiefel manifold.

To this end, we replace the Euclidean gradient by an approximate gradient mapping for orthogonality-constrained optimization [41]. For each agent, define

$$H_i(X) = G_i(X) + \beta X(X^\top X - I_r),$$

where

$$G_i(X) = \nabla f_i(XX^\top X) \frac{3I_r - X^\top X}{2} - X \text{sym}(X^\top \nabla f_i(XX^\top X)).$$

The first term provides an ambient descent surrogate tailored to the geometry of the Stiefel manifold, while the penalty term controls orthogonality violation. This construction allows the geometric information to enter through a local map H_i without changing the linear communication structure of EXTRA. For the averaged analysis, we also use the corresponding global map

$$H(X) = G(X) + \beta X(X^\top X - I_r),$$

where G is defined from $f := n^{-1} \sum_{i=1}^n f_i$ in the same way as G_i is defined from f_i . After introducing the algorithm, we write

$$\mathbf{H}_k := [H_1(X_{1,k})^\top, \dots, H_n(X_{n,k})^\top]^\top, \quad \bar{H}_k := \frac{1}{n} \sum_{i=1}^n H_i(X_{i,k}).$$

The resulting algorithm, called RF-EXTRA, is stated in Algorithm 1. The initialization follows the EXTRA mechanism and uses the local map H_i evaluated at the initial point.

In stacked form, Algorithm 1 reads

$$\mathbf{X}_{k+1} = \mathbf{W}\mathbf{X}_k + \mathbf{s}_k, \tag{2.3}$$

$$\mathbf{s}_{k+1} = \mathbf{s}_k + (\mathbf{W} - \mathbf{V})\mathbf{X}_k - \alpha(\mathbf{H}_{k+1} - \mathbf{H}_k). \tag{2.4}$$

Averaging (2.3)–(2.4) and using the doubly stochasticity of $\mathbf{W}$ and $\mathbf{V}$ yields

$$\bar{\mathbf{s}}_k = -\alpha \bar{H}_k, \quad \bar{\mathbf{X}}_{k+1} = \bar{\mathbf{X}}_k - \alpha \bar{H}_k.$$

This averaged recursion is the key link between RF-EXTRA and the descent analysis developed in Section 3.

RF-EXTRA has two main advantages. First, it preserves the EXTRA mechanism, so the communication layer remains linear and fully decentralized. Second, it avoids explicit retractions at every iteration. Each step requires one neighbor mixing update for $\mathbf{X}_k$, one evaluation of the local surrogate map H_i , and one auxiliary correction update for $\mathbf{s}_k$. Consequently, the method replaces repeated orthogonalization with a cheaper ambient-space computation while still maintaining effective control of feasibility and consensus through the penalty term and the correction state.

Algorithm 1 RF-EXTRA

Require: Mixing matrix W , correction matrix $V = \theta I_n + (1 - \theta)W$ with $\theta \in (0, 1/2]$, stepsize $\alpha > 0$, penalty parameter $\beta > 0$, initial matrices $X_{i,0} \in \mathbb{R}^{d \times r}$ for $i = 1, \dots, n$

- 1: **for** each agent $i = 1, \dots, n$ **do**
- 2: Compute $H_i(X_{i,0})$
- 3: Set $s_{i,0} = -\alpha H_i(X_{i,0})$
- 4: **end for**
- 5: **for** $k = 0, 1, 2, \dots$ **do**
- 6: **for** each agent $i = 1, \dots, n$ in parallel **do**
- 7: Exchange $X_{j,k}$ with neighbors j
- 8: Update the primal variable:

$$X_{i,k+1} = \sum_{j=1}^n w_{ij} X_{j,k} + s_{i,k}$$

- 9: Compute $H_i(X_{i,k+1})$
- 10: Update the auxiliary variable:

$$s_{i,k+1} = s_{i,k} + \sum_{j=1}^n (w_{ij} - v_{ij}) X_{j,k} - \alpha (H_i(X_{i,k+1}) - H_i(X_{i,k}))$$

- 11: **end for**
- 12: **end for**

3 Convergence Analysis

This section establishes convergence of the averaged RF-EXTRA iterate. The analysis combines two ingredients: descent properties inherited from the centralized surrogate analysis on the Stiefel manifold and a contractive recursion for the joint error. The latter is handled in an S-norm, which is essential for the EXTRA subsystem.

3.1 Standing assumptions

Before stating the assumptions, we collect the constants used throughout the analysis. Let

$$b(X) := \frac{1}{4} \|X^\top X - I_r\|_F^2, \quad \mathcal{R} := \{X \in \mathbb{R}^{d \times r} : \|X^\top X - I_r\|_F \leq 1/6\},$$

and define the explicit bounded set

$$\mathcal{B} := \{X \in \mathbb{R}^{d \times r} : \|X\|_F \leq \sqrt{7r/6} + 1\}.$$

We use L_f for a Lipschitz constant of ∇f on $\{XX^\top X: X \in \mathcal{B}\}$, and we define

$$L_g := \sup_{X \neq Y \in \mathcal{B}} \frac{\|G(X) - G(Y)\|_F}{\|X - Y\|_F}, \quad L_b := \sup_{X \neq Y \in \mathcal{B}} \frac{\|\nabla b(X) - \nabla b(Y)\|_F}{\|X - Y\|_F},$$

$$M_g := \sup_{X \in \mathcal{R}} \|G(X)\|_F, \quad C_0 := \sup_{X \in \mathcal{R}} \|\nabla f(XX^\top X)\|_F,$$

and the shorthand

$$L_H := L_g + \beta L_b, \quad M_H := \sup_{X \in \mathcal{R}} \|H(X)\|_F.$$

We collect the assumptions used in the convergence analysis.

Assumption 1 (Smooth local models). Each local objective f_i is smooth, and the constants $L_f, L_g, L_b, M_g, C_0, L_H$, and M_H defined above are finite on the indicated sets.

Assumption 2 (Network). The mixing matrix W is symmetric and doubly stochastic, the communication graph is connected, and the second largest singular value satisfies $\sigma_2(W) < 1$.

3.2 Averaged-sequence ingredients

The next lemma records the two properties imported from the centralized analysis: coercivity of the surrogate map on the neighborhood of interest and one-step propagation of the averaged iterate inside that neighborhood.

Lemma 3.1 (Averaged neighborhood preservation). *If*

$$\beta \geq \frac{6+21C_0}{5},$$

then, for every $X \in \mathcal{R}$,

$$\|H(X)\|_F^2 \geq \|G(X)\|_F^2 + \beta \|X^\top X - I_r\|_F^2. \quad (3.1)$$

Moreover, if

$$\bar{X}_k \in \mathcal{R}, \quad \|\mathbf{x}_k - \bar{\mathbf{x}}_k\|_F \leq \min \left\{ 1, \frac{\sqrt{n}}{L_H} \right\}, \quad \alpha \leq \min \left\{ \frac{1}{4L_H}, \frac{1}{4\beta L_b}, \frac{1}{4\beta} \right\},$$

and

$$\beta \geq 12\sqrt{2}(M_g + 1),$$

then

$$\bar{X}_{k+1} \in \mathcal{R}.$$

Proof. The coercivity estimate (3.1) is exactly the statement of DESTINY Lemma 2 for the same pointwise map

$$H(X) = G(X) + \beta X(X^\top X - I_r)$$

on the same region $\mathcal{R}$; hence it may be cited directly.

For the second claim, let

$$Q(X) := X^\top X - I_r, \quad b(X) := \frac{1}{4} \|Q(X)\|_F^2.$$

Then $\nabla b(X) = XQ(X)$. As before, if

$$J := \frac{1}{n} \mathbf{1}_n \mathbf{1}_n^\top \otimes I_d,$$

then $J\mathbf{W} = J$, $J\mathbf{V} = J$, and $J(\mathbf{W} - \mathbf{V}) = 0$. Applying J to the auxiliary recursion gives

$$J\mathbf{s}_{k+1} = J\mathbf{s}_k - \alpha J(\mathbf{H}_{k+1} - \mathbf{H}_k).$$

Because $s_{i,0} = -\alpha H_i(X_{i,0})$, averaging yields $\bar{s}_0 = -\alpha \bar{H}_0$, and induction gives

$$\bar{s}_k = -\alpha \bar{H}_k \quad \text{for all } k \geq 0.$$

Applying J to the primal recursion,

$$\mathbf{X}_{k+1} = \mathbf{W}\mathbf{X}_k + \mathbf{s}_k,$$

and using $J\mathbf{W} = J$, we obtain

$$\bar{X}_{k+1} = \bar{X}_k + \bar{s}_k = \bar{X}_k - \alpha \bar{H}_k.$$

Now define

$$\zeta_k := H(\bar{X}_k) - \bar{H}_k - G(\bar{X}_k).$$

Since $H(\bar{X}_k) = G(\bar{X}_k) + \beta \nabla b(\bar{X}_k)$, the averaged recursion becomes

$$\bar{X}_{k+1} = \bar{X}_k - \alpha \beta \nabla b(\bar{X}_k) + \alpha \zeta_k. \quad (3.2)$$

Moreover,

$$\|X_{i,k} - \bar{X}_k\|_F \leq \|\mathbf{X}_k - \bar{\mathbf{X}}_k\|_F \leq 1, \quad i = 1, \dots, n.$$

Since $\bar{X}_k \in \mathcal{R}$, we have $\|\bar{X}_k^\top \bar{X}_k - I_r\|_F \leq 1/6$, and hence

$$\|\bar{X}_k\|_F^2 = \text{tr}(\bar{X}_k^\top \bar{X}_k) \leq \text{tr}(I_r) + \sqrt{r} \|\bar{X}_k^\top \bar{X}_k - I_r\|_F \leq r + \frac{\sqrt{r}}{6} \leq \frac{7r}{6}.$$

Therefore $\|\bar{X}_k\|_F \leq \sqrt{7r/6}$, hence $X_{i,k} \in \mathcal{B}$ for all i . Therefore H_i is L_H -Lipschitz at the points used below, and

$$\|\zeta_k\|_F \leq \|H(\bar{X}_k) - \bar{H}_k\|_F + \|G(\bar{X}_k)\|_F \leq \frac{L_H}{\sqrt{n}} \|\mathbf{X}_k - \bar{\mathbf{X}}_k\|_F + M_g \leq 1 + M_g. \quad (3.3)$$

Because $\bar{X}_k \in \mathcal{R}$, we have $\|Q(\bar{X}_k)\|_F \leq 1/6$, so every eigenvalue of $\bar{X}_k^\top \bar{X}_k$ lies in $[5/6, 7/6]$. We now estimate the feasibility of $\bar{X}_{k+1}$ directly through the singular values of $\bar{X}_k$. Let

$$t := \alpha\beta, \quad \bar{X}_k = U\Sigma V^\top, \quad q_j := \sigma_j^2 - 1,$$

where $\Sigma = \text{diag}(\sigma_j)$. Since $\bar{X}_k \in \mathcal{R}$, we have $|q_j| \leq 1/6$. Define the penalty-only point

$$Z_k := \bar{X}_k - t\nabla b(\bar{X}_k).$$

Using $\nabla b(\bar{X}_k) = U\Sigma(\Sigma^2 - I_r)V^\top$, we have

$$Z_k = U\text{diag}(\sigma_j(1 - tq_j))V^\top.$$

Therefore

$$\begin{aligned} \|Z_k^\top Z_k - I_r\|_F^2 &= \sum_j ((1 + q_j)(1 - tq_j)^2 - 1)^2 \\ &= \sum_j q_j^2 (1 - 2t(1 + q_j) + t^2 q_j(1 + q_j))^2 \\ &\leq \left(1 - \frac{5}{3}t\right)^2 \sum_j q_j^2 \\ &= \left(1 - \frac{5}{3}t\right)^2 \|\bar{X}_k^\top \bar{X}_k - I_r\|_F^2. \end{aligned} \tag{3.4}$$

Indeed, the scalar factor in the second line is bounded by $1 - 5t/3$ for $q_j \in [-1/6, 1/6]$ and $t \leq 1/4$: if $q_j \geq 0$, then $1 - 2t(1 + q_j) + t^2 q_j(1 + q_j) \leq 1 - 2t \leq 1 - 5t/3$; if $q_j < 0$, then $1 - 2t(1 + q_j) + t^2 q_j(1 + q_j) \leq 1 - 2t(5/6) = 1 - 5t/3$.

It remains to include the perturbation $\alpha\tilde{\xi}_k$ in (3.2). Since the singular values of Z_k are $\sigma_j|1 - tq_j|$, the bounds $|q_j| \leq 1/6$ and $t \leq 1/4$ imply $\|Z_k\|_2 \leq \sqrt{7/6}$. Hence, using $\bar{X}_{k+1} = Z_k + \alpha\tilde{\xi}_k$,

$$\begin{aligned} \|\bar{X}_{k+1}^\top \bar{X}_{k+1} - I_r\|_F &\leq \|Z_k^\top Z_k - I_r\|_F + 2\alpha\|Z_k\|_2\|\tilde{\xi}_k\|_F + \alpha^2\|\tilde{\xi}_k\|_F^2 \\ &\leq \left(1 - \frac{5}{3}t\right)\|\bar{X}_k^\top \bar{X}_k - I_r\|_F + 2t\sqrt{\frac{7}{6}}\frac{\|\tilde{\xi}_k\|_F}{\beta} + t^2\left(\frac{\|\tilde{\xi}_k\|_F}{\beta}\right)^2. \end{aligned}$$

Combine the assumed β condition and (3.3), we have

$$\frac{\|\tilde{\xi}_k\|_F}{\beta} \leq \frac{1}{12\sqrt{2}}.$$

Since $\|\bar{X}_k^\top \bar{X}_k - I_r\|_F \leq 1/6$, $t \leq 1/4$, and

$$2\sqrt{\frac{7}{6}}\frac{1}{12\sqrt{2}} + \frac{t}{288} \leq \frac{1}{6} + \frac{1}{1152} \leq \frac{5}{18},$$

we obtain

$$\|\bar{X}_{k+1}^\top \bar{X}_{k+1} - I_r\|_F \leq \left(1 - \frac{5}{3}t\right)\frac{1}{6} + \frac{5t}{18} = \frac{1}{6}.$$

Hence $\bar{X}_{k+1} \in \mathcal{R}$, as claimed. $\square$

3.3 Joint-error recursion in an equivalent norm

We next control the disagreement-correction subsystem associated with RF-EXTRA. The key point is that one should not work with the Frobenius operator norm of the linear transition matrix. We therefore begin with a standard equivalent-norm lemma.

A second subtlety is that the Lipschitz estimate for $H_i(X_{i,k+1}) - H_i(X_{i,k})$ involves the next local iterate. Accordingly, the recursion below is proved in a one-step bootstrap form: averaged-region control together with small primal and auxiliary disagreement first imply $X_{i,k}, X_{i,k+1} \in \mathcal{B}$, and only then is the L_H -Lipschitz property invoked.

Lemma 3.2 ([46]). *For any given $\varepsilon > 0$, there exists an operator norm $\|\cdot\|_S$ (dependent on A and ε) such that $\|A\|_S \leq \rho(A) + \varepsilon$. Consequently, if $\rho(A) < 1$, then there exists an operator norm $\|\cdot\|_S$ for which $\|A\|_S < 1$.*

Under Assumption 2, define

$$P := \begin{bmatrix} \mathbf{W} - J & I \\ \mathbf{W} - \mathbf{V} & I - J \end{bmatrix}.$$

Using the standard state-space representation of EXTRA [4, 5], the RF-EXTRA disagreement state satisfies

$$\mathbf{p}_{k+1} = P\mathbf{p}_k + \alpha \delta_k, \quad \delta_k := \begin{bmatrix} 0 \\ (J - I)(\mathbf{H}_{k+1} - \mathbf{H}_k) \end{bmatrix}. \quad (3.5)$$

By [47], the condition $\sigma_2(W) < 1$ implies $\rho(P) < 1$. Applying Lemma 3.2 to $A = P$, we may therefore fix an operator norm $\|\cdot\|_S$ and constants $c_1, c_2 > 0$ such that

$$\|P\|_S =: \sigma < 1, \quad c_1 \|A\|_F \leq \|A\|_S \leq c_2 \|A\|_F$$

for all stacked variables A in the joint-error space.

Lemma 3.3 (Joint-error recursion in an equivalent norm). *Recall the stacked quantities $\mathbf{p}_k$, $\mathbf{H}_k$, and σ introduced earlier. Let $c_1, c_2 > 0$ satisfy*

$$c_1 \|A\|_F \leq \|A\|_S \leq c_2 \|A\|_F.$$

Let

$$a := \frac{1 + \sigma}{2}, \quad B_H := c_2 L_H \sqrt{n}.$$

Assume that, for some $\delta_S > 0$,

$$\begin{aligned} \bar{X}_k \in \mathcal{R}, \quad \|\mathbf{p}_k\|_S \leq \delta_S, \quad \delta_S \leq c_1 \min \left\{ \frac{1}{1 + \sigma_2(W)}, \frac{\sqrt{n}}{L_H} \right\}, \\ \alpha \leq \min \left\{ \frac{1}{4L_H}, \frac{1}{4\beta L_b}, \frac{1}{4\beta}, \frac{c_1(1 - \sigma)}{8c_2 L_H}, 1 \right\}, \quad \beta \geq 12\sqrt{2}(M_g + 1). \end{aligned}$$

Then $\bar{X}_{k+1} \in \mathcal{R}$, and

$$\|\mathbf{p}_{k+1}\|_S \leq a \|\mathbf{p}_k\|_S + \alpha^2 B_H \|H(\bar{X}_k)\|_F. \quad (3.6)$$

Proof. Let

$$\tilde{\mathbf{X}}_k := \mathbf{X}_k - \bar{\mathbf{X}}_k, \quad \tilde{\mathbf{s}}_k := \mathbf{s}_k - \bar{\mathbf{s}}_k.$$

From (3.5), we have

$$\|\mathbf{p}_{k+1}\|_S \leq \sigma \|\mathbf{p}_k\|_S + \alpha \|\delta_k\|_S. \quad (3.7)$$

By norm equivalence and the bound $\|\mathbf{p}_k\|_S \leq \delta_S$,

$$\|\tilde{\mathbf{X}}_k\|_F \leq \|\mathbf{p}_k\|_F \leq \frac{\delta_S}{c_1}, \quad \|\tilde{\mathbf{s}}_k\|_F \leq \|\mathbf{p}_k\|_F \leq \frac{\delta_S}{c_1}.$$

Since $\bar{\mathbf{X}}_k \in \mathcal{R}$, we have $\|\bar{\mathbf{X}}_k^\top \bar{\mathbf{X}}_k - I_r\|_F \leq 1/6$, and hence

$$\|\bar{\mathbf{X}}_k\|_F^2 = \text{tr}(\bar{\mathbf{X}}_k^\top \bar{\mathbf{X}}_k) \leq \text{tr}(I_r) + \sqrt{r} \|\bar{\mathbf{X}}_k^\top \bar{\mathbf{X}}_k - I_r\|_F \leq r + \frac{\sqrt{r}}{6} \leq \frac{7r}{6}.$$

Therefore $\|\bar{\mathbf{X}}_k\|_F \leq \sqrt{7r/6}$. Together with $\delta_S/c_1 \leq 1/(1+\sigma_2(W)) \leq 1$, this yields

$$\|X_{i,k}\|_F \leq \|\bar{\mathbf{X}}_k\|_F + \|X_{i,k} - \bar{\mathbf{X}}_k\|_F \leq \sqrt{7r/6} + \|\tilde{\mathbf{X}}_k\|_F \leq \sqrt{7r/6} + 1, \quad i = 1, \dots, n,$$

hence $X_{i,k} \in \mathcal{B}$ for all i .

The assumptions imply $\|\mathbf{X}_k - \bar{\mathbf{X}}_k\|_F \leq \delta_S/c_1 \leq \min\{1, \sqrt{n}/L_H\}$, and hence all conditions in the second part of Lemma 3.1 are satisfied. Therefore

$$\bar{\mathbf{X}}_{k+1} \in \mathcal{R}.$$

Moreover, subtracting the average from the primal recursion yields

$$\tilde{\mathbf{X}}_{k+1} = (\mathbf{W} - J)\tilde{\mathbf{X}}_k + \tilde{\mathbf{s}}_k.$$

Therefore

$$\|\tilde{\mathbf{X}}_{k+1}\|_F \leq \sigma_2(W) \|\tilde{\mathbf{X}}_k\|_F + \|\tilde{\mathbf{s}}_k\|_F \leq (1 + \sigma_2(W)) \frac{\delta_S}{c_1} \leq 1.$$

Using again $\bar{\mathbf{X}}_{k+1} \in \mathcal{R}$, we obtain $\|\bar{\mathbf{X}}_{k+1}\|_F \leq \sqrt{7r/6}$ and hence

$$\|X_{i,k+1}\|_F \leq \|\bar{\mathbf{X}}_{k+1}\|_F + \|X_{i,k+1} - \bar{\mathbf{X}}_{k+1}\|_F \leq \sqrt{7r/6} + \|\tilde{\mathbf{X}}_{k+1}\|_F \leq \sqrt{7r/6} + 1, \quad i = 1, \dots, n,$$

so $X_{i,k+1} \in \mathcal{B}$ for all i . Thus every point at which H_i is evaluated in the present step lies in $\mathcal{B}$. Since $J(\mathbf{W} - \mathbf{V}) = 0$ and $\bar{\mathbf{s}}_k = -\alpha(\mathbf{1}_n \otimes I_d)\bar{H}_k$, the primal increment satisfies

$$\mathbf{X}_{k+1} - \mathbf{X}_k = (\mathbf{W} - I)\tilde{\mathbf{X}}_k + \tilde{\mathbf{s}}_k - \alpha\bar{\mathbf{H}}_k,$$

where $\bar{H}_k := n^{-1} \sum_{i=1}^n H_i(X_{i,k})$ and $\bar{\mathbf{H}}_k := (\mathbf{1}_n \otimes I_d)\bar{H}_k$. Therefore,

$$\|\mathbf{X}_{k+1} - \mathbf{X}_k\|_F \leq 2\|\tilde{\mathbf{X}}_k\|_F + \|\tilde{\mathbf{s}}_k\|_F + \alpha\|\bar{\mathbf{H}}_k\|_F. \quad (3.8)$$

Because $X_{i,k}, X_{i,k+1} \in \mathcal{B}$ for every i , each H_i is L_H -Lipschitz along this step, and therefore

$$\|H(\bar{X}_k) - \bar{H}_k\|_F \leq \frac{L_H}{n} \sum_{i=1}^n \|\bar{X}_k - X_{i,k}\|_F \leq \frac{L_H}{\sqrt{n}} \|\mathbf{X}_k - \bar{\mathbf{X}}_k\|_F,$$

which implies

$$\|\bar{\mathbf{H}}_k\|_F \leq L_H \|\tilde{\mathbf{X}}_k\|_F + \sqrt{n} \|H(\bar{X}_k)\|_F. \quad (3.9)$$

Substituting (3.9) into (3.8) and using $\alpha L_H \leq 1$ give

$$\|\mathbf{X}_{k+1} - \mathbf{X}_k\|_F \leq 3\|\tilde{\mathbf{X}}_k\|_F + \|\tilde{\mathbf{s}}_k\|_F + \alpha\sqrt{n} \|H(\bar{X}_k)\|_F \leq 4\|\mathbf{p}_k\|_F + \alpha\sqrt{n} \|H(\bar{X}_k)\|_F.$$

Therefore we have

$$\|\mathbf{H}_{k+1} - \mathbf{H}_k\|_F \leq L_H \|\mathbf{X}_{k+1} - \mathbf{X}_k\|_F \leq 4L_H \|\mathbf{p}_k\|_F + \alpha L_H \sqrt{n} \|H(\bar{X}_k)\|_F.$$

Hence

$$\|\delta_k\|_S \leq c_2 \|\mathbf{H}_{k+1} - \mathbf{H}_k\|_F \leq \frac{4c_2 L_H}{c_1} \|\mathbf{p}_k\|_S + \alpha B_H \|H(\bar{X}_k)\|_F.$$

Combining this with (3.7) yields

$$\|\mathbf{p}_{k+1}\|_S \leq \left(\sigma + \frac{4c_2 L_H}{c_1} \alpha \right) \|\mathbf{p}_k\|_S + \alpha^2 B_H \|H(\bar{X}_k)\|_F.$$

The restriction $\alpha \leq c_1(1-\sigma)/(8c_2 L_H)$ gives

$$\sigma + \frac{4c_2 L_H}{c_1} \alpha \leq \frac{1+\sigma}{2} = a,$$

which proves (3.6). $\square$

Remark. A direct Frobenius-norm contraction argument is unavailable for the RF-EXTRA transition matrix P . Indeed, if $u \neq 0$ satisfies $Ju=0$ and $z=[0;u]^\top$, then $Pz=[u;u]^\top$, so $\|Pz\|_F = \sqrt{2}\|u\|_F > \|z\|_F$. Thus any contraction inequality directly under $\|\cdot\|_F$ is infeasible. This is why the introduction of the S -norm is crucial.

3.4 Neighborhood propagation

Once the joint-error recursion is available, we propagate the invariant neighborhood generated by Lemma 3.4. The pointwise boundedness facts needed later are simple consequences of the already-established disagreement control and averaged-region control, so we use them directly inside the final theorem proof instead of isolating them as a separate lemma.

Lemma 3.4 (One-step propagation of the joint neighborhood). *Let $\delta_S > 0$, and define*

$$\mathcal{N}_S(\delta_S) := \{(\mathbf{X}, \mathbf{s}) : \|\mathbf{p}\|_S \leq \delta_S\}.$$

Let

$$B_H := c_2 L_H \sqrt{n}, \quad a := \frac{1+\sigma}{2}.$$

Assume the hypotheses of Lemma 3.3. If, for some $k \geq 0$,

$$\bar{X}_k \in \mathcal{R}, \quad (\mathbf{X}_k, \mathbf{s}_k) \in \mathcal{N}_S(\delta_S), \quad \alpha \leq \sqrt{\frac{(1-a)\delta_S}{B_H M_H}},$$

then

$$\bar{X}_{k+1} \in \mathcal{R}, \quad (\mathbf{X}_{k+1}, \mathbf{s}_{k+1}) \in \mathcal{N}_S(\delta_S).$$

Proof. By Lemma 3.3, we have $\bar{X}_{k+1} \in \mathcal{R}$ and

$$\|\mathbf{p}_{k+1}\|_S \leq a\|\mathbf{p}_k\|_S + \alpha^2 B_H \|H(\bar{X}_k)\|_F.$$

Since $\bar{X}_k \in \mathcal{R}$ and $\mathcal{R}$ is bounded, continuity of H implies

$$\|H(\bar{X}_k)\|_F \leq M_H.$$

Using $\|\mathbf{p}_k\|_S \leq \delta_S$, we obtain

$$\|\mathbf{p}_{k+1}\|_S \leq a\delta_S + \alpha^2 B_H M_H.$$

The stepsize restriction yields

$$\alpha^2 B_H M_H \leq (1-a)\delta_S,$$

hence

$$\|\mathbf{p}_{k+1}\|_S \leq a\delta_S + (1-a)\delta_S = \delta_S.$$

Hence $(\mathbf{X}_{k+1}, \mathbf{s}_{k+1}) \in \mathcal{N}_S(\delta_S)$. □

The recursive bound above also yields a summed estimate for the joint error, which will be used in the final descent argument.

Lemma 3.5 (Summed joint-error bound). *Assume that, for each $k = 0, \dots, K$, the one-step hypotheses of Lemma 3.3 hold. Then*

$$\sum_{k=0}^K \left(\|\mathbf{X}_{k+1} - \bar{\mathbf{X}}_{k+1}\|_F^2 + \|\mathbf{s}_{k+1} - \bar{\mathbf{s}}_{k+1}\|_F^2 \right) \leq C_{P,0} + \alpha^2 C_{P,1} \sum_{k=0}^K \|H(\bar{X}_k)\|_F^2, \quad (3.10)$$

where

$$C_{P,0} := \frac{2a^2}{c_1^2(1-a^2)} \|\mathbf{p}_0\|_S^2, \quad C_{P,1} := \frac{2nc_2^2 L_H^2}{c_1^2(1-a)^2}.$$

Proof. Unrolling (3.6) gives

$$\|\mathbf{p}_{k+1}\|_S \leq a^{k+1} \|\mathbf{p}_0\|_S + \alpha^2 B_H \sum_{t=0}^k a^{k-t} \|H(\bar{X}_t)\|_F.$$

Set $h_t := \|H(\bar{X}_t)\|_F$. Using $(u+v)^2 \leq 2u^2 + 2v^2$, we obtain

$$\|\mathbf{p}_{k+1}\|_S^2 \leq 2a^{2k+2} \|\mathbf{p}_0\|_S^2 + 2\alpha^4 B_H^2 \left(\sum_{t=0}^k a^{k-t} h_t \right)^2.$$

Summing over $k=0, \dots, K$ yields

$$\sum_{k=0}^K \|\mathbf{p}_{k+1}\|_S^2 \leq 2 \|\mathbf{p}_0\|_S^2 \sum_{k=0}^K a^{2k+2} + 2\alpha^4 B_H^2 \sum_{k=0}^K \left(\sum_{t=0}^k a^{k-t} h_t \right)^2.$$

The geometric-series bound gives

$$\sum_{k=0}^K a^{2k+2} \leq \frac{a^2}{1-a^2}.$$

For the second term, Cauchy–Schwarz implies

$$\left(\sum_{t=0}^k a^{k-t} h_t \right)^2 \leq \left(\sum_{t=0}^k a^{k-t} \right) \left(\sum_{t=0}^k a^{k-t} h_t^2 \right) \leq \frac{1}{1-a} \sum_{t=0}^k a^{k-t} h_t^2.$$

Summing over k and exchanging the order of summation,

$$\sum_{k=0}^K \left(\sum_{t=0}^k a^{k-t} h_t \right)^2 \leq \frac{1}{1-a} \sum_{t=0}^K h_t^2 \sum_{k=t}^K a^{k-t} \leq \frac{1}{(1-a)^2} \sum_{t=0}^K h_t^2.$$

Therefore,

$$\sum_{k=0}^K \|\mathbf{p}_{k+1}\|_S^2 \leq \frac{2a^2}{1-a^2} \|\mathbf{p}_0\|_S^2 + \frac{2\alpha^4 B_H^2}{(1-a)^2} \sum_{k=0}^K \|H(\bar{X}_k)\|_F^2.$$

Finally, norm equivalence gives

$$\|\mathbf{X}_{k+1} - \bar{\mathbf{X}}_{k+1}\|_F^2 + \|\mathbf{s}_{k+1} - \bar{\mathbf{s}}_{k+1}\|_F^2 \leq \frac{1}{c_1^2} \|\mathbf{p}_{k+1}\|_S^2,$$

and, since $\alpha \leq 1$ and $B_H = c_2 L_H \sqrt{n}$,

$$\sum_{k=0}^K \left(\|\mathbf{X}_{k+1} - \bar{\mathbf{X}}_{k+1}\|_F^2 + \|\mathbf{s}_{k+1} - \bar{\mathbf{s}}_{k+1}\|_F^2 \right) \leq \frac{2a^2}{c_1^2(1-a^2)} \|\mathbf{p}_0\|_S^2 + \alpha^2 \frac{2nc_2^2 L_H^2}{c_1^2(1-a)^2} \sum_{k=0}^K \|H(\bar{X}_k)\|_F^2.$$

This is exactly (3.10). $\square$

3.5 Exact $\mathcal{O}(1/K)$ convergence rate

We now combine averaged descent with the summed joint-error bound. Summing the descent inequality along the averaged trajectory and comparing the different terms through Lemma 3.5 yields an exact $\mathcal{O}(1/K)$ convergence rate. Recall that the descent coefficient c_H is defined in Proposition 3.6.

Lemma 3.6 (Descent lemma for the averaged RF-EXTRA sequence). *Let $\bar{H}_k := n^{-1} \sum_{i=1}^n H_i(X_{i,k})$, let $Q_k := \bar{X}_k^\top \bar{X}_k - I_r$. Suppose Assumption 1 holds, suppose moreover that $X_{i,k} \in \mathcal{B}$ for all $i = 1, \dots, n$ and $\bar{X}_k \in \mathcal{R}$. Assume further that the remaining hypotheses in the second part of Lemma 3.1, including its parameter assumptions, are satisfied at iteration k . Assume $\beta \geq 56L_f^2$, and let $\alpha \leq \min \left\{ \frac{3}{8L_h}, \frac{1}{4L_H} \right\}$, then we have*

$$h(\bar{X}_{k+1}) \leq h(\bar{X}_k) - \frac{3\alpha}{16} \|H(\bar{X}_k)\|_F^2 + \Gamma_X(\alpha) \|\mathbf{X}_k - \bar{\mathbf{X}}_k\|_F^2, \quad (3.11)$$

where

$$\Gamma_X(\alpha) := \frac{17\alpha L_H^2}{8n} + \frac{\alpha^2 L_h L_H^2}{n} \leq \Gamma_X^\# := \frac{5L_H}{8n}.$$

Proof. Write $H_k := H(\bar{X}_k)$. By the second part of Lemma 3.1, we obtain $\bar{X}_{k+1} \in \mathcal{R}$. Hence $\bar{X}_k, \bar{X}_{k+1} \in \mathcal{R} \subset \mathcal{B}$, and h is L_h -smooth on $\mathcal{B}$,

$$h(\bar{X}_{k+1}) \leq h(\bar{X}_k) - \alpha \langle \nabla h(\bar{X}_k), \bar{H}_k \rangle + \frac{\alpha^2 L_h}{2} \|\bar{H}_k\|_F^2. \quad (3.12)$$

Split the inner product into

$$-\alpha \langle \nabla h(\bar{X}_k) - H_k, \bar{H}_k \rangle - \alpha \langle H_k, \bar{H}_k \rangle.$$

Since $\bar{X}_k \in \mathcal{R}$ implies $\|\bar{X}_k\|_2^2 \leq \frac{7}{6}$, we have

$$\|\nabla h(\bar{X}_k) - H_k\|_F^2 \leq \frac{9}{4} \|\bar{X}_k(\bar{X}_k^\top \bar{X}_k - I_r)\|_F^2 \leq \frac{21}{8} L_f^2 \|Q_k\|_F^2.$$

Moreover,

$$\|H_k - \bar{H}_k\|_F \leq \frac{L_H}{\sqrt{n}} \|\mathbf{X}_k - \bar{\mathbf{X}}_k\|_F, \quad \|\bar{H}_k\|_F^2 \leq 2\|H_k\|_F^2 + \frac{2L_H^2}{n} \|\mathbf{X}_k - \bar{\mathbf{X}}_k\|_F^2.$$

Next we bound $\langle \nabla h(\bar{X}_k) - H_k, \bar{H}_k \rangle$, $\langle H_k, \bar{H}_k \rangle$ and $\frac{\alpha^2 L_h}{2} \|\bar{H}_k\|_F^2$ separately. For the first one, Young's inequality gives

$$\begin{aligned} -\alpha \langle \nabla h(\bar{X}_k) - H_k, \bar{H}_k \rangle &\leq \alpha \left(4\|\nabla h(\bar{X}_k) - H_k\|_F^2 + \frac{1}{16} \|\bar{H}_k\|_F^2 \right) \\ &\leq \alpha \left[4 \cdot \frac{21}{8} L_f^2 \|Q_k\|_F^2 + \frac{1}{16} \left(2\|H_k\|_F^2 + \frac{2L_H^2}{n} \|\mathbf{X}_k - \bar{\mathbf{X}}_k\|_F^2 \right) \right] \\ &= \frac{21}{2} \alpha L_f^2 \|Q_k\|_F^2 + \frac{\alpha}{8} \|H_k\|_F^2 + \frac{\alpha L_H^2}{8n} \|\mathbf{X}_k - \bar{\mathbf{X}}_k\|_F^2. \end{aligned}$$

For the second term, we write

$$\begin{aligned}
 -\alpha \langle H_k, \bar{H}_k \rangle &= -\alpha \langle H_k, \bar{H}_k - H_k \rangle - \alpha \|H_k\|_F^2 \\
 &\leq \alpha \left(\frac{1}{8} \|H_k\|_F^2 + 2 \|\bar{H}_k - H_k\|_F^2 \right) - \alpha \|H_k\|_F^2 \\
 &\leq \alpha \left(\frac{1}{8} \|H_k\|_F^2 + \frac{2L_H^2}{n} \|\mathbf{X}_k - \bar{\mathbf{X}}_k\|_F^2 \right) - \alpha \|H_k\|_F^2 \\
 &= -\frac{7\alpha}{8} \|H_k\|_F^2 + \frac{2\alpha L_H^2}{n} \|\mathbf{X}_k - \bar{\mathbf{X}}_k\|_F^2.
 \end{aligned}$$

Finally, using the bound on $\|\bar{H}_k\|_F^2$, we obtain

$$\begin{aligned}
 \frac{\alpha^2 L_h}{2} \|\bar{H}_k\|_F^2 &\leq \frac{\alpha^2 L_h}{2} \left(2 \|H_k\|_F^2 + \frac{2L_H^2}{n} \|\mathbf{X}_k - \bar{\mathbf{X}}_k\|_F^2 \right) \\
 &= \alpha^2 L_h \|H_k\|_F^2 + \frac{\alpha^2 L_h L_H^2}{n} \|\mathbf{X}_k - \bar{\mathbf{X}}_k\|_F^2.
 \end{aligned}$$

Therefore, plugging the above three expressions into (3.12), we have

$$h(\bar{X}_{k+1}) \leq h(\bar{X}_k) + \frac{21}{2} \alpha L_f^2 \|Q_k\|_F^2 + \Gamma_X(\alpha) \|\mathbf{X}_k - \bar{\mathbf{X}}_k\|_F^2 + \left(-\frac{3\alpha}{4} + \alpha^2 L_h \right) \|H_k\|_F^2.$$

Because $\alpha \leq 3/(8L_h)$, the last coefficient is bounded above by $-3\alpha/8$. Since $\bar{X}_k \in \mathcal{R}$, Lemma 3.1 gives

$$\|H_k\|_F^2 \geq \beta \|Q_k\|_F^2,$$

hence

$$\frac{21}{2} \alpha L_f^2 \|Q_k\|_F^2 \leq \frac{21L_f^2}{2\beta} \alpha \|H_k\|_F^2.$$

Substituting this estimate yields

$$h(\bar{X}_{k+1}) \leq h(\bar{X}_k) - \left(\frac{3}{8} - \frac{21L_f^2}{2\beta} \right) \alpha \|H_k\|_F^2 + \Gamma_X(\alpha) \|\mathbf{X}_k - \bar{\mathbf{X}}_k\|_F^2,$$

Because $\beta \geq 56L_f^2$, the coefficient satisfies $\frac{3}{8} - \frac{21L_f^2}{2\beta} \geq \frac{3}{16}$, which gives (3.11). Since $\alpha \leq \min \left\{ \frac{3}{8L_h}, \frac{1}{4L_H} \right\}$, then

$$\Gamma_X(\alpha) \leq \frac{17\alpha L_H^2}{8n} + \frac{3\alpha L_H^2}{8n} = \frac{5\alpha L_H^2}{2n} \leq \frac{5L_H}{8n} = \Gamma_X^\sharp.$$

□

Theorem 3.1 (Exact $\mathcal{O}(1/K)$ convergence rate for the averaged iterate). *Assume Assumptions 1 and 2. Let $\delta_S > 0$ satisfy*

$$\delta_S \leq c_1 \min \left\{ \frac{1}{1+\sigma_2(W)}, \frac{\sqrt{n}}{L_H} \right\},$$

and suppose $\bar{X}_0 \in \mathcal{R}$ and $(\mathbf{X}_0, \mathbf{s}_0) \in \mathcal{N}_S(\delta_S)$. Assume that β satisfies

$$\beta \geq \max \left\{ 56L_f^2, \frac{6+21C_0}{5}, 12\sqrt{2}(M_g+1) \right\}.$$

Let

$$\begin{aligned} \Delta_h &:= h(\bar{X}_0) - \inf_{X \in \mathcal{R}} h(X), \quad \Gamma_X^\sharp := \frac{5L_H}{8n}, \quad C_{P,0}^\sharp := \|\mathbf{X}_0 - \bar{\mathbf{X}}_0\|_F^2 + C_{P,0}, \\ B_H &:= c_2 L_H \sqrt{n}, \quad a := \frac{1+\sigma}{2}. \end{aligned}$$

If

$$\alpha \leq \bar{\alpha} := \min \left\{ \frac{1}{4L_H}, \frac{1}{4\beta L_b}, \frac{1}{4\beta}, \frac{c_1(1-\sigma)}{8c_2 L_H}, \sqrt{\frac{(1-a)\delta_S}{B_H M_H}}, \frac{1}{M_H+1}, \frac{3}{8L_h}, \frac{3}{32\Gamma_X^\sharp C_{P,1}}, 1 \right\}, \quad (3.13)$$

then, for every integer $K \geq 0$,

$$\frac{1}{K+1} \sum_{k=0}^K \|H(\bar{X}_k)\|_F^2 \leq \frac{32(\Delta_h + \Gamma_X^\sharp C_{P,0}^\sharp)}{3\alpha(K+1)}. \quad (3.14)$$

Consequently,

$$\frac{1}{K+1} \sum_{k=0}^K \|G(\bar{X}_k)\|_F^2 \leq \frac{32(\Delta_h + \Gamma_X^\sharp C_{P,0}^\sharp)}{3\alpha(K+1)}. \quad (3.15)$$

$$\frac{1}{K+1} \sum_{k=0}^K \|\bar{X}_k^\top \bar{X}_k - I_r\|_F^2 \leq \frac{32(\Delta_h + \Gamma_X^\sharp C_{P,0}^\sharp)}{3\alpha\beta(K+1)}. \quad (3.16)$$

Hence the RF-EXTRA averaged iterate attains an exact $\mathcal{O}(1/K)$ convergence rate.

Proof. Repeated application of Lemma 3.4 gives $\bar{X}_k \in \mathcal{R}$ and $(\mathbf{X}_k, \mathbf{s}_k) \in \mathcal{N}_S(\delta_S)$ for all $k \geq 0$. Hence $\bar{X}_{k+1} \in \mathcal{R}$ as well. By norm equivalence,

$$\|\mathbf{X}_k - \bar{\mathbf{X}}_k\|_F \leq \|\mathbf{p}_k\|_F \leq \frac{1}{c_1} \|\mathbf{p}_k\|_S \leq \frac{\delta_S}{c_1} \leq \frac{1}{1+\sigma_2(W)} \leq 1.$$

Since $\bar{X}_k \in \mathcal{R}$, we have $\|\bar{X}_k\|_F \leq \sqrt{7r/6}$; therefore, for every i ,

$$\|X_{i,k}\|_F \leq \|\bar{X}_k\|_F + \|X_{i,k} - \bar{X}_k\|_F \leq \sqrt{7r/6} + \|\mathbf{X}_k - \bar{\mathbf{X}}_k\|_F \leq \sqrt{7r/6} + 1,$$

so $X_{i,k} \in \mathcal{B}$ for all i, k . Moreover, $\mathcal{R} \subset \mathcal{B}$, hence $\bar{X}_k, \bar{X}_{k+1} \in \mathcal{B}$. Therefore Lemma 3.6 applies at every step. Summing (3.11) from $k=0$ to K and using $\Gamma_X(\alpha) \leq \Gamma_X^\sharp$ yield

$$h(\bar{X}_{K+1}) \leq h(\bar{X}_0) - \frac{3\alpha}{16} \sum_{k=0}^K \|H(\bar{X}_k)\|_F^2 + \Gamma_X^\sharp \sum_{k=0}^K \|\mathbf{X}_k - \bar{\mathbf{X}}_k\|_F^2.$$

Now split the consensus sum as

$$\sum_{k=0}^K \|\mathbf{X}_k - \bar{\mathbf{X}}_k\|_F^2 = \|\mathbf{X}_0 - \bar{\mathbf{X}}_0\|_F^2 + \sum_{k=0}^{K-1} \|\mathbf{X}_{k+1} - \bar{\mathbf{X}}_{k+1}\|_F^2.$$

Applying Lemma 3.5 to the second term and discarding the nonnegative $\|\mathbf{s}_{k+1} - \bar{\mathbf{s}}_{k+1}\|_F^2$ part give

$$\sum_{k=0}^K \|\mathbf{X}_k - \bar{\mathbf{X}}_k\|_F^2 \leq C_{P,0}^\sharp + \alpha^2 C_{P,1} \sum_{k=0}^{K-1} \|H(\bar{X}_k)\|_F^2 \leq C_{P,0}^\sharp + \alpha^2 C_{P,1} \sum_{k=0}^K \|H(\bar{X}_k)\|_F^2.$$

Substituting this estimate gives

$$h(\bar{X}_{K+1}) \leq h(\bar{X}_0) - \left(\frac{3\alpha}{16} - \Gamma_X^\sharp \alpha^2 C_{P,1} \right) \sum_{k=0}^K \|H(\bar{X}_k)\|_F^2 + \Gamma_X^\sharp C_{P,0}^\sharp.$$

By (3.13),

$$\Gamma_X^\sharp \alpha C_{P,1} \leq \frac{3}{32}, \quad \text{hence} \quad \frac{3\alpha}{16} - \Gamma_X^\sharp \alpha^2 C_{P,1} \geq \frac{3\alpha}{32}.$$

Since $h(\bar{X}_{K+1}) \geq \inf_{X \in \mathcal{R}} h(X)$, we conclude that

$$\frac{3\alpha}{32} \sum_{k=0}^K \|H(\bar{X}_k)\|_F^2 \leq h(\bar{X}_0) - h(\bar{X}_{K+1}) + \Gamma_X^\sharp C_{P,0}^\sharp \leq \Delta_h + \Gamma_X^\sharp C_{P,0}^\sharp.$$

Dividing by $K+1$ proves (3.14). Finally, Lemma 3.1 gives

$$\|H(\bar{X}_k)\|_F^2 \geq \|G(\bar{X}_k)\|_F^2 + \beta \|\bar{X}_k^\top \bar{X}_k - I_r\|_F^2,$$

hence (3.15) and (3.16) follow immediately from (3.14). $\square$

The convergence proof preserves the averaged iterate and the joint error introduced earlier, while replacing a Frobenius-norm contraction argument by an equivalent-norm analysis of the EXTRA subsystem. As a result, the final estimate is obtained through a transparent descent-and-accumulation argument: averaged descent provides the main decrease, the joint-error recursion quantifies the network-induced perturbation, and the summed joint-error bound makes it possible to conclude under explicit conditions on α and β .

4 Experiments

In this section, we compare RF-EXTRA with four decentralized baselines: DPRGD [29], DPRGT [30], DESTINY [41], and REXTRA [31]. Our implementation follows the experimental setting of [31] as closely as possible.

4.1 Decentralized principal component analysis

The decentralized principal component analysis (PCA) problem seeks a common low-dimensional subspace that preserves the maximal variation of data distributed across multiple agents. It can be formulated as

$$\min_{X \in \mathcal{M}^n} -\frac{1}{2n} \sum_{i=1}^n \text{tr}(X_i^\top A_i^\top A_i X_i), \quad \text{s.t.} \quad X_1 = \dots = X_n, \quad (4.1)$$

where $\mathcal{M}^n := \text{St}(d, r)^n$, $\text{St}(d, r)$ is the Stiefel manifold, n is the number of agents, and $A_i \in \mathbb{R}^{m_i \times d}$ is the local data matrix stored at agent i . As in [31], if X^* solves (4.1), then X^*Q is also a solution for any orthogonal matrix $Q \in \mathbb{R}^{r \times r}$. We therefore measure the distance to a reference solution by

$$d_s(X, X^*) := \min_{Q^\top Q = QQ^\top = I_r} \|XQ - X^*\|.$$

4.1.1 Synthetic dataset

We fix $m_1 = \dots = m_n = 1000$, $d = 10$, $r = 5$, and $n = 8$. A Gaussian matrix $B \in \mathbb{R}^{1000n \times d}$ is generated and decomposed as $B = U\Sigma V^\top$. We then set $\tilde{\Sigma} = \text{diag}(\tilde{\xi}^j)$ with $\tilde{\xi} \in (0, 1)$ and construct $A = U\tilde{\Sigma}V^\top$, whose rows are partitioned uniformly across the n agents. The first r columns of V form a solution to (4.1). In all experiments we use $\tilde{\xi} = 0.8$. For the PCA problem, the effective scaling is chosen as

$$\alpha = \frac{\hat{\beta}n}{\sum_{i=1}^n m_i}.$$

All algorithms use constant step sizes. For the synthetic comparison below, the step size is selected from

$$\{1, 2, 4, 6, 8\} \times \{10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}\}.$$

We first study the robustness of RF-EXTRA with respect to graph topology and the internal parameter β . Figure 1 shows that the same qualitative behavior is preserved across ring, star, and ER graphs of different densities, and also across a wide range of internal β values. This supports the claim that the retraction-free recursion is reasonably stable under moderate changes in both network structure and internal tuning.

We next compare the wall-clock efficiency of RF-EXTRA and REXTRA under matched graph settings. Table 1 shows that, once both methods are stably tuned, RF-EXTRA is

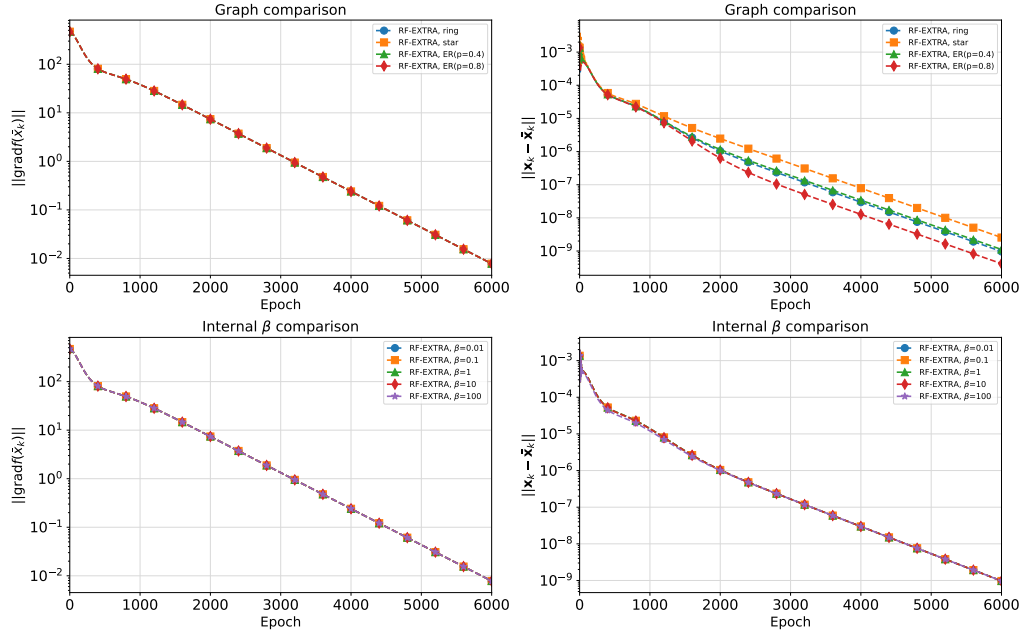


Figure 1: Synthetic decentralized PCA: robustness of RF-EXTRA with respect to graph topology and internal β . Top row: stationarity and consensus trajectories under ring, star, ER(0.4), and ER(0.8) graphs. Bottom row: the same quantities when the internal parameter β varies in $\{0.01, 0.1, 1, 10, 100\}$ on the ring graph.

typically slightly faster than REXTRA, which is consistent with the additional cost of retraction operations in the latter.

After these focused robustness and efficiency checks, we turn to the standard synthetic decentralized PCA task under the same ER(0.6) graph, seed, and epoch budget for all methods. Figure 2 reports the best run of RF-EXTRA, DESTINY, DPRGT, DPRGD, and REXTRA under this unified search space. RF-EXTRA is competitive with the strongest baselines across the communication budget, and its performance is close to that of REXTRA. Combined with the wall-clock comparison in Table 1, this indicates that RF-EXTRA achieves similar empirical behavior while avoiding the extra retraction cost.

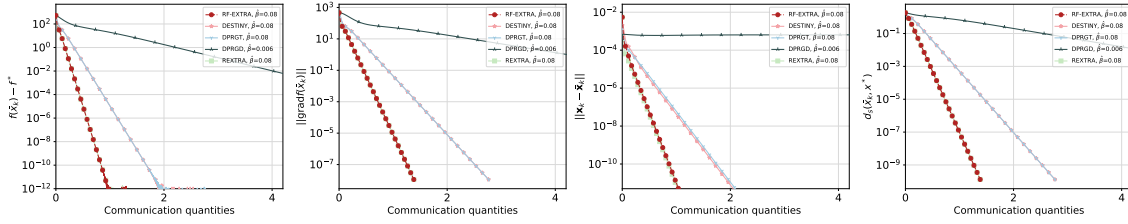
4.1.2 MNIST dataset

To further evaluate the practical behavior of RF-EXTRA on real data, we conduct experiments on the MNIST dataset. The dataset contains 60,000 handwritten digit images of size 28×28 , which are used to generate the local matrices A_i . We first normalize the pixel values by dividing by 255 and then randomly partition the data into $n = 8$ agents with equal numbers of samples. As a result, each agent holds a local matrix A_i of dimension $\frac{60000}{n} \times 784$. We compute the top $r = 2$ principal components with ambient dimension $d = 784$, use a stopping threshold $\|\text{grad} f(\tilde{x}_k)\| < 10^{-6}$, and choose the effective scaling

$$\alpha = \frac{\hat{\beta}}{60000}.$$

Table 1: Final wall-clock time (seconds) of RF-EXTRA and REXTRA on ER graphs. Smaller is better.

Method / stepsize	$p=0.4$	$p=0.6$	$p=0.8$
RF-EXTRA, $\hat{\beta}=0.3$	0.617	0.583	0.588
RF-EXTRA, $\hat{\beta}=0.05$	3.388	3.268	3.384
RF-EXTRA, $\hat{\beta}=0.008$	20.433	20.716	22.598
REXTRA, $\hat{\beta}=0.3$	0.609	0.645	0.670
REXTRA, $\hat{\beta}=0.05$	3.613	3.574	3.570
REXTRA, $\hat{\beta}=0.008$	23.447	22.364	23.669

Figure 2: Synthetic decentralized PCA on ER(0.6) versus communication quantities. Each method uses its best step size selected from $\{1,2,4,6,8\} \times \{10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}\}$. Under this matched search space, RF-EXTRA, DESTINY, DPRGT, and REXTRA all select $\hat{\beta}=0.08$, while DPRGD selects $\hat{\beta}=0.006$.

Similar to the synthetic PCA setting, all methods use constant step sizes, and we select the best available step size from the grid

$$\{1,2,6\} \times \{10^{-4}, 10^{-3}, 10^{-2}\}.$$

Among the runs already completed in this grid, the strongest available setting is $\hat{\beta}=0.06$ for RF-EXTRA, DESTINY, and DPRGT, while the best available DPRGD run uses $\hat{\beta}=0.02$. Accordingly, Figure 3 reports RF-EXTRA, DESTINY, and DPRGT at $\hat{\beta}=0.06$ together with DPRGD at $\hat{\beta}=0.02$.

Figure 3 shows that RF-EXTRA and DESTINY perform similarly and both improve on DPRGT in the later stage, while DPRGD remains clearly weaker within the same communication budget.

4.2 Decentralized low-rank matrix completion

Low-rank matrix completion (LRMC) aims to recover a low-rank matrix $A \in \mathbb{R}^{d \times T}$ from a subset of its observed entries. Let Ω denote the index set of observed entries. The rank- r LRMC problem can be written as

$$\min_{X \in \text{Gr}(d,r), V \in \mathbb{R}^{r \times T}} \frac{1}{2} \|\mathcal{P}_{\Omega} \odot (XV - A)\|^2,$$

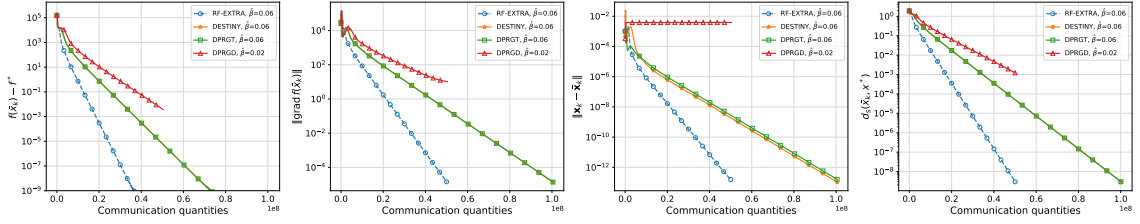


Figure 3: Decentralized PCA on the MNIST dataset versus communication quantities. RF-EXTRA, DESTINY, and DPRGT are plotted at $\hat{\beta}=0.06$, while DPRGD is plotted at its best existing MNIST noMPI step size $\hat{\beta}=0.02$ among the currently available runs from $\{1,2,6\} \times \{10^{-4}, 10^{-3}, 10^{-2}\}$.

where $\text{Gr}(d, r)$ is the Grassmann manifold, $\odot$ denotes the Hadamard product, and $\mathcal{P}_\Omega$ is the observation mask. In the decentralized setting, the partially observed matrix is partitioned column-wise across n agents into local blocks $A_1, \dots, A_n$. Using the standard representation of the Grassmann manifold through the Stiefel manifold, the problem can be reformulated as a decentralized optimization problem over $\text{St}(d, r)$,

$$\min_{X_i \in \text{St}(d, r)} \frac{1}{2} \sum_{i=1}^n \|\mathcal{P}_{\Omega_i} \odot (X_i V_i(X_i) - A_i)\|^2, \quad \text{s.t.} \quad X_1 = \dots = X_n,$$

where Ω_i is the local observation pattern and $V_i(X_i)$ is the corresponding least-squares factor. Following [31], we consider a synthetic setup with $T = 1000$, $d = 100$, $r = 5$, and $n = 8$, where the ground-truth low-rank matrix is generated from Gaussian factors and the observation mask is sampled with rate $\mu = r(d + T - r) / (dT)$. More precisely, if $L \in \mathbb{R}^{d \times r}$ and $R \in \mathbb{R}^{r \times T}$ are Gaussian factors, then the target matrix is generated as

$$A = LR + 10^{-3}E,$$

where $E \in \mathbb{R}^{d \times T}$ has i.i.d. standard normal entries. Thus, the LRMC instance is mildly perturbed from an exactly rank- r model. We adopt a ring graph to model the communication network among agents. For LRMC, we use the effective scaling

$$\alpha = \hat{\beta} \times n.$$

All algorithms are implemented with constant step sizes, and the candidate step sizes are selected from the grid

$$\{1, 2, 5, 8\} \times \{10^{-4}, 10^{-3}, 10^{-2}\}.$$

We run each method for at most 1500 epochs and terminate early if $\|\text{grad} f(\bar{x}_k)\| < 10^{-6}$.

Because the perturbed LRMC instance does not admit a practically meaningful exact reference solution, we only report the stationarity and Euclidean consensus panels in the revised communication plot. Figure 4 shows that RF-EXTRA reaches low stationarity and consensus levels more rapidly than DESTINY and DPRGT in the plotted regime, while DPRGD remains less competitive within the same budget. Figure 5 further compares representative RF-EXTRA and DESTINY step sizes on the same communication axis and shows the same qualitative trend.

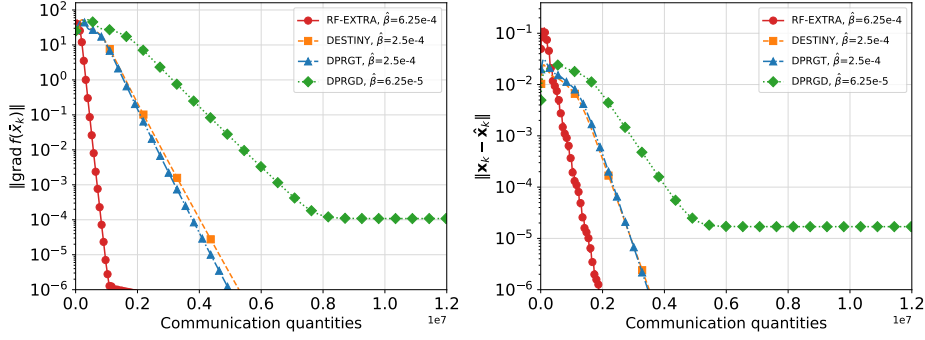


Figure 4: Decentralized LRMC on the ring graph versus communication quantities. Only the stationarity and Euclidean consensus panels are shown, since the perturbed instance does not provide an exact-solution reference panel of comparable value. For each method, the step size is selected by the earliest epoch at which $\|\text{grad} f(\bar{x}_k)\| < 10^{-6}$.

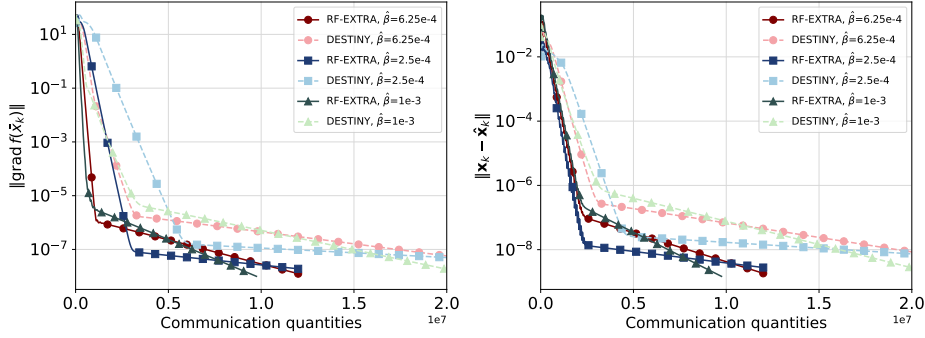


Figure 5: Decentralized LRMC on the ring graph versus communication quantities for representative RF-EXTRA and DESTINY step sizes. We again retain only the stationarity and Euclidean consensus panels.

5 Conclusion

We proposed RF-EXTRA, a decentralized retraction-free method for optimization on the Stiefel manifold. The method combines an ambient-space surrogate for the orthogonality-constrained local models with an EXTRA-based primal-dual recursion, thereby preserving a simple decentralized communication structure while avoiding explicit per-iteration retractions.

On the theoretical side, the analysis is built on the averaged iterate and the joint error $(X_k - \bar{X}_k, s_k - \bar{s}_k)$. By establishing a contractive recursion for the joint error under an equivalent norm and then comparing the different terms in the descent analysis, we obtain an exact $\mathcal{O}(1/K)$ convergence guarantee for the averaged iterate.

The experiments indicate that RF-EXTRA is a competitive retraction-free decentralized solver for optimization problems on the Stiefel manifold. On PCA and low-rank matrix completion, it delivers favorable empirical performance and communication efficiency relative to the reported baselines. These results support retraction-free decen-

tralized correction as a promising direction for large-scale optimization on matrix manifolds.

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